SIMATS ENGINEERING

CONTINOUS INTERNAL ASSESSMENT- CONCEPT MAPPING

COURSE NAME: DATABASE MANAGEMENT SYSTEMS
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Questions:05 **Total marks:100**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Hamshalakshmi K (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ANNEXURE



CONCEPT MAPPING

QUESTIONS		
Q.No	Question	Bloom's Level
1	A university database contains Persons, Students, Faculty, and Staff, where specialized entities inherit common properties from a general entity. Explain how the EER model extends the basic ER model using generalization, specialization, and aggregation. Construct a concept map illustrating these concepts and their relationships.	BL2 – Understand
2	An online banking application receives a user's request to retrieve account information. Construct a visual concept map explaining the flow of the request through the DBMS software architecture, including the query processor, transaction manager, storage manager, database files, and data retrieval process.	BL2 – Understand
3	Different organizations use different data models according to their requirements. Construct a concept map connecting Hierarchical, Network, Relational, and ER/EER models with suitable real-world applications such as organizational systems, transportation networks, banking systems, and university databases. Explain the suitability of each model.	BL2 – Understand
4	A university database undergoes changes in its logical structure and physical storage without changing how users access the data. Construct a concept map explaining schema, instance, logical schema, physical schema, logical data independence, and physical data independence, and show the relationship among these concepts.	BL2 – Understand
5	An organization uses a DBMS to manage its business data. Construct a concept map showing the roles of the Database Administrator (DBA), End User, and Application Developer. Include their major responsibilities and explain how they interact within the overall DBMS environment.	BL1 – Remember

MAPPING

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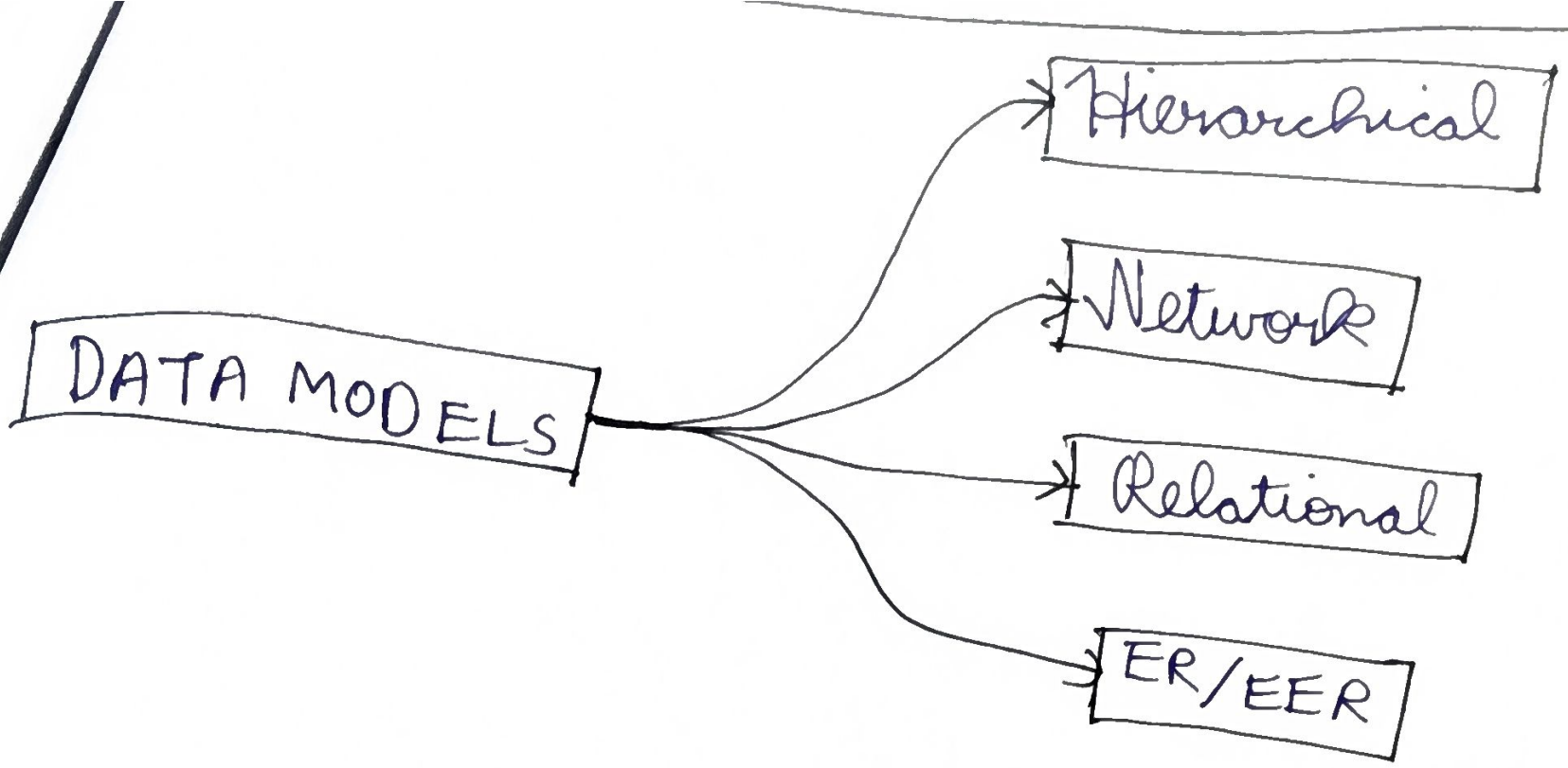
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ERR MODEL

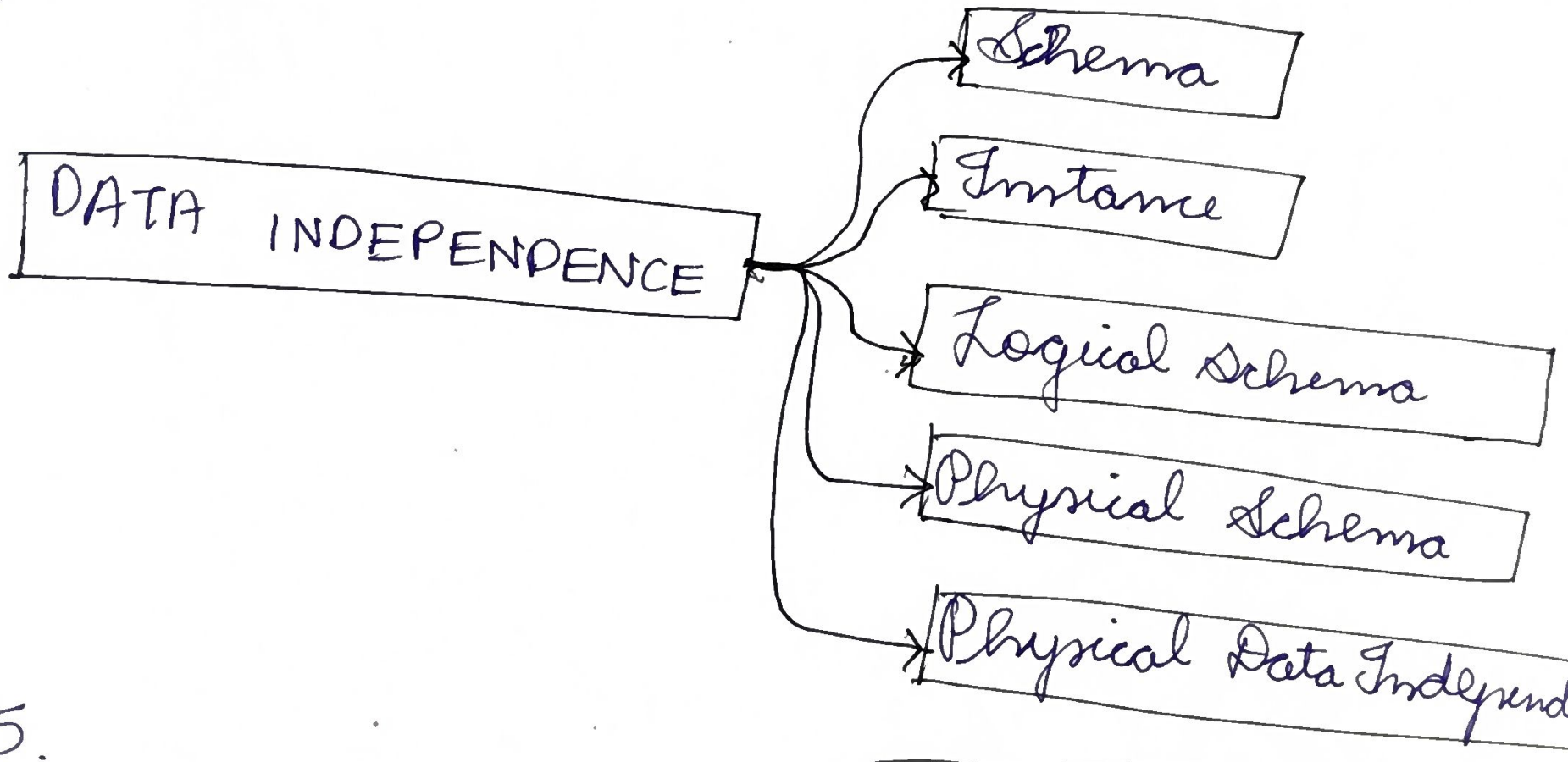
- Person, Students, Staff, Faculty
- Specialization
- Aggregation
- Super class / Subclass
- Inheritance of common places

DBMS ARCHITECTURE

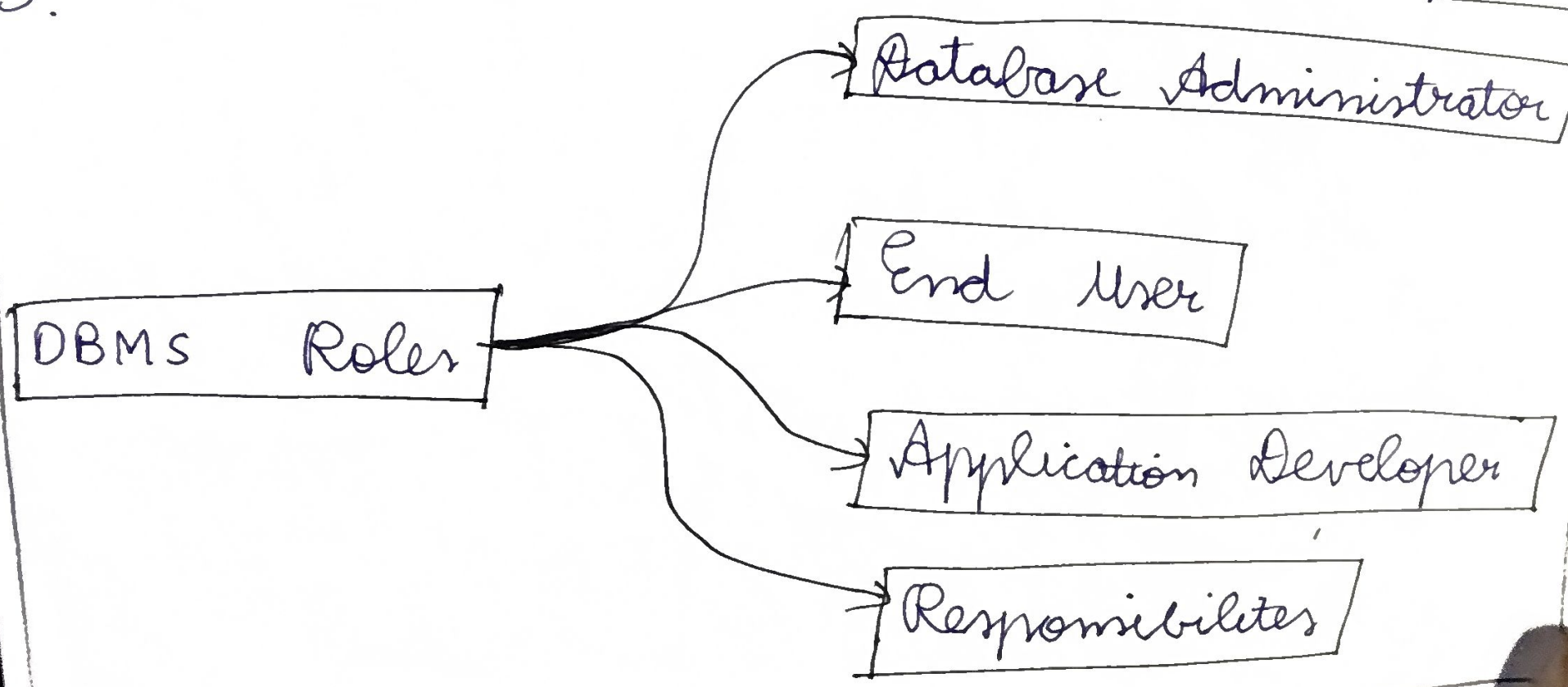
- User Request
- Query Processor
- Transaction Manager
- Storage Manager
- Database files
- Data Retrival


DATA MODELS
Hierarchical
Network
Relational
ER/EER

4.


DATA INDEPENDENCE
Schema
Instance
Logical Schema
Physical Schema
Physical Data Independan

5.


DBMS Roles
Database Administrator
End User
Application Developer
Responsibilites